

Enterprise Infrastructure Plan

Company: Yoma Maintenance Inc.

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Role: Junior System Administrator

PART 1: Company Profile

- **Company Name:** Yoma Maintenance Inc.
- **Nature of Business:** Software and Website Maintenance Services
- **Company Vision:** To be the most trusted technical partner for businesses globally by ensuring their websites and software applications run flawlessly 24/7.
- **Company Mission:** To provide proactive software updates, website monitoring, and bug-fixing services that eliminate downtime and secure our clients' digital assets.
- **Office Location:** 12th Floor, Cyber Sigma, BGC, Taguig City
- **Organizational Structure & Employee Distribution:** 20 Employees Total
 - Information Technology: 5 Employees
 - Human Resources: 4 Employees
 - Finance: 5 Employees
 - Sales: 6 Employees

PART 2: Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computers	9	HR (4), Finance (5)	Standard workstations for HR and Finance operations.
HW-002	Laptops	11	IT (5), Sales (6)	Mobile computing for coding (IT) and client presentations (Sales).
HW-003	Server	1	IT	Centralized application hosting and domain controller.
HW-004	Router	1	IT	Main gateway for managing local network traffic and internet connection.
HW-005	Switch (24-port)	1	IT	To connect all wired devices (desktops, printers, server).
HW-006	Printer	2	HR/Finance, Sales	Shared network printing for document processing.
HW-007	UPS	10	All	Battery backup for the 9 Desktops and 1 Server.
HW-008	Wireless Access Point	2	All	Wi-Fi coverage for the entire office floor.
HW-009	NAS Storage	1	IT	Centralized network storage for file sharing and backups.
HW-010	External Backup Drive	2	IT	Cold storage and

				offline backup copies of critical company data.
HW-011	Monitors	14	All	9 for Desktops, 5 as secondary extended monitors for IT Laptops.

PART 3: Enterprise Software Inventory

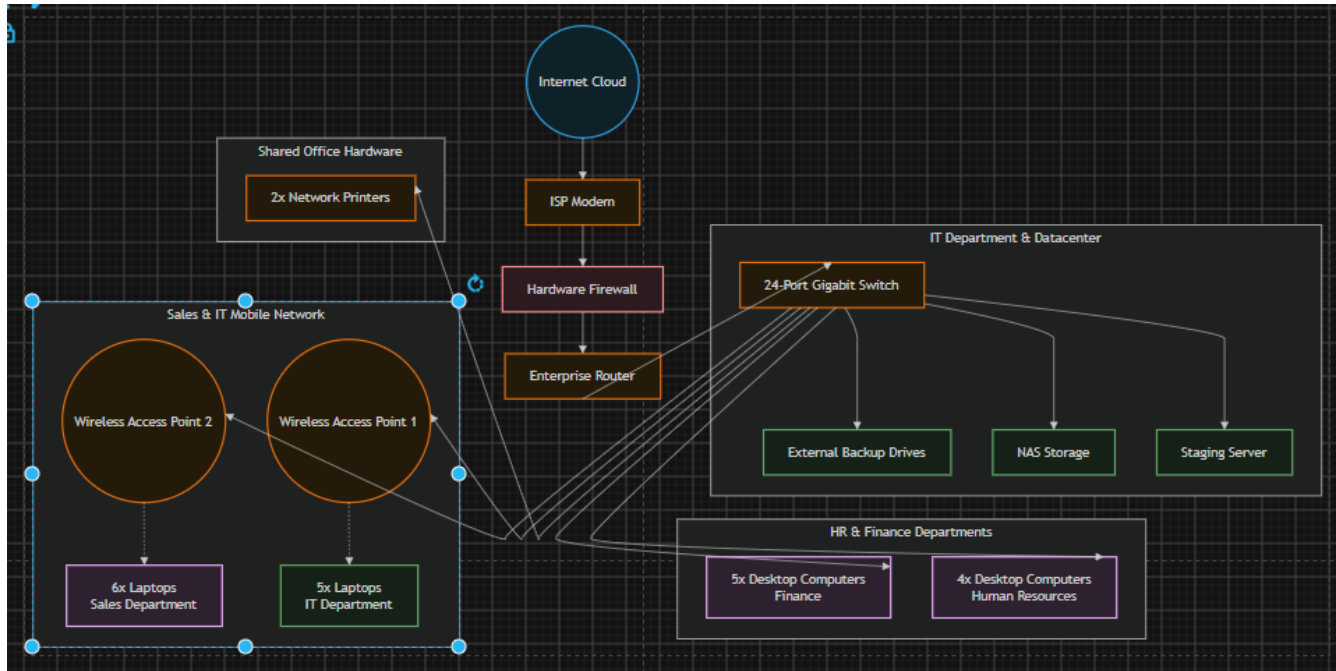
Software	Version	License	Purpose
Windows 11 Pro	22H2	Commercial	Standard OS for all employee desktops and laptops, allowing domain join.
Ubuntu Server	22.04 LTS	Open-Source	Secure, lightweight OS for hosting internal web applications.
Microsoft Office	365 Business	Subscription	For creating documents, spreadsheets, and emails (HR, Finance, Sales).
VS Code	Latest	Open-Source	Primary code editor used by the IT software development team.
Git	Latest	Open-Source	Version control system for tracking source code changes.
GitHub Desktop	Latest	Open-Source	GUI for Git to manage code repositories easily.
VirtualBox	7.0	Open-Source	For IT to test applications in isolated virtual environments before deployment.
Google Chrome	Latest	Freeware	Standard web browser for internet access and web apps.
Microsoft Defender	Built-in	Commercial	Essential antivirus and threat protection for all endpoints.
AnyDesk	Latest	Commercial	Remote desktop tool for IT to troubleshoot employee computers remotely.
7-Zip	Latest	Open-Source	File archiver for compressing and extracting large project files.

PART 4: Enterprise Network Inventory

Item	Quantity	Purpose
ISP Modem	1	Provided by ISP to connect the office to the internet.
Router	1	To route traffic between the local network and the internet.
Firewall (Hardware)	1	To monitor and block malicious incoming and outgoing network traffic.
Switch (24-Port Gigabit)	1	Core switch to connect all Ethernet cables from the wall ports.
Access Point	2	To provide wireless connectivity for laptops and mobile devices.

Patch Panel (24-Port)	1	To organize and terminate all network cables coming from office walls.
CAT6 Cables (Boxes)	2	High-speed cabling for wired connections across the office floor.
RJ45 Connectors (Pack)	1	Connectors required to terminate the ends of the CAT6 cables.

PART 5: Enterprise Network Diagram



PART 6: System Administration Roles

- **Helpdesk Technician:**
- **Responsibilities:** Provides first-line technical support to employees, resets passwords, and fixes basic hardware/software issues.
- **Skills:** Customer service, basic troubleshooting, OS configuration.
- **Common Tools:** Ticketing systems (Zendesk), AnyDesk, Active Directory Users & Computers.
- **Certifications:** CompTIA A+, Microsoft 365 Certified Fundamentals.
- **Network Administrator:**
- **Responsibilities:** Designs, configures, and maintains the physical and wireless network infrastructure to ensure uptime.
- **Skills:** IP subnetting, routing protocols, firewall configuration.
- **Common Tools:** Wireshark, Cisco Packet Tracer, PuTTY, Ping/Traceroute.
- **Certifications:** CCNA (Cisco Certified Network Associate), CompTIA Network+.
- **Linux System Administrator:**
- **Responsibilities:** Deploys, secures, and updates Linux-based servers (like Ubuntu Server) hosting company applications.
- **Skills:** Command-line proficiency, Bash scripting, server security management.
- **Common Tools:** SSH, Vim/Nano, htop, systemd.
- **Certifications:** RHCSA (Red Hat Certified System Administrator), Linux+.

- **Cloud Administrator:**
- **Responsibilities:** Manages off-site cloud infrastructure, cloud backups, and virtual machines on platforms like AWS or Azure.
- **Skills:** Cloud resource provisioning, IAM (Identity and Access Management), cost optimization.
- **Common Tools:** Terraform, AWS CLI, Azure Portal.
- **Certifications:** AWS Certified SysOps Administrator, Microsoft Certified: Azure Administrator Associate.

How They Work Together:

For a maintenance company like Yoma Maintenance, these four roles are crucial for service delivery. When a client reports a website issue, the Helpdesk Technician logs the ticket. If the issue is related to our internal connectivity preventing our developers from accessing the client's site, the Network Administrator steps in. The Linux System Administrator manages our internal staging servers where we test the website bug fixes. Finally, the Cloud Administrator ensures that the client's live websites and their weekly backups on platforms like AWS are correctly configured and highly available.

PART 7: Infrastructure Recommendations

- **Internet Provider:** I recommend acquiring a dedicated Enterprise Fiber connection (e.g., PLDT Enterprise or Converge ICT) with at least 500 Mbps symmetrical speeds. Since our business relies on maintaining clients' digital assets online, a strict Service Level Agreement (SLA) for 99.9% uptime is non-negotiable.
- **Server Specifications:** For our on-premise staging server, a rack-mounted server with an Intel Xeon processor, 64GB ECC RAM, and 4TB of SSD storage configured in RAID 1 is recommended. This will allow us to locally host heavy virtual machines to replicate client website environments for safe debugging.
- **Backup Strategy:** Implement the 3-2-1 Backup Rule: Maintain 3 total copies of data, across 2 different media (NAS and External Backup Drive), with 1 copy stored offsite or in the cloud.
- **Security Recommendations:**
- **Antivirus:** Microsoft Defender for Endpoint should be strictly managed via group policy, ensuring all employee machines are free from malware that could accidentally be uploaded to a client's website.
- **Password Policy:** Enforce a minimum 12-character password length, mandatory Multi-Factor Authentication (MFA) for all accounts, and a 90-day password expiration policy to secure access to client credentials.
- **Expansion Plan:** As Yoma Maintenance grows and handles larger enterprise clients, we should invest heavily in automated cloud monitoring tools (like Datadog or New Relic) and shift our staging environments fully to AWS or Azure to handle heavier workloads without buying more physical servers.

PART 8: Personal Reflection

Through this enterprise infrastructure planning project, I gained a comprehensive understanding of the meticulous preparation required before a single piece of equipment is purchased. I learned that designing an IT infrastructure for a maintenance company like Yoma Maintenance requires focusing heavily on reliability, secure storage for client backups, and flexible workstations for the development team.

The most challenging task was formulating the exact quantities and justifying the hardware and software inventories. Ensuring that the 20 employees across the IT, HR, Finance, and Sales departments had the appropriate tools—such as assigning laptops for mobility to Sales and IT, and desktops for stationary roles in HR and Finance—required careful logical mapping. Additionally, visualizing the physical connections of the firewall, switch, and access points for the network diagram tested my understanding of how data flows securely within a corporate environment.

Planning is absolutely critical before deployment because it prevents costly mistakes, ensures hardware compatibility, and establishes a clear roadmap for security. Without a proper plan, an organization risks purchasing incompatible software or leaving glaring security vulnerabilities that could compromise both our data and our clients' websites.

This project will significantly help me become a better System Administrator because it shifted my perspective from a purely technical mindset to a business-oriented one. I now understand that a System Administrator's role starts at the drawing board, ensuring the infrastructure directly supports the company's goal of maintaining excellence for its clients.